```
package payment;

public abstract class Payment {

    protected double amount;

    public Payment(double amount) {
        if (!Double.isFinite(amount) || amount <= 0) {
            throw new IllegalArgumentException(
                "Amount must be positive and finite.");
        }
        this.amount = amount;
    }

    public abstract void pay();

    // Overloaded pay() methods
    public void pay(String reference) {
        System.out.println("Reference: " + reference);
        pay();
    }

    public void pay(double discountPercent, String reference) {
        if (!Double.isFinite(discountPercent)
                || discountPercent < 0 || discountPercent > 100) {
            throw new IllegalArgumentException("Invalid discount.");
        }

        double finalAmount = amount * (1 - discountPercent / 100.0);
        System.out.println("Reference: " + reference);
        System.out.printf("Discounted amount: Rs. %.2f%n", finalAmount);
    }
}
```

```
package payment;

public interface Refundable {
    void refund(double amount);
}
```

```
package payment;

public class CardPayment extends Payment implements Refundable {

    public CardPayment(double amount) {
        super(amount);
    }

    @Override
    public void pay() {
        System.out.printf("Paid Rs. %.2f using Card.%n", amount);
    }

    @Override
    public void refund(double refundAmount) {
        if (!Double.isFinite(refundAmount)
                || refundAmount <= 0 || refundAmount > amount) {
            throw new IllegalArgumentException("Invalid refund amount.");
        }
        System.out.printf("Card refund: Rs. %.2f%n", refundAmount);
    }
}



package payment;

public class UPIPayment extends Payment implements Refundable {

    public UPIPayment(double amount) {
        super(amount);
    }

    @Override
    public void pay() {
        System.out.printf("Paid Rs. %.2f using UPI.%n", amount);
    }

    @Override
    public void refund(double refundAmount) {
        if (!Double.isFinite(refundAmount)
                || refundAmount <= 0 || refundAmount > amount) {
```

```
            throw new IllegalArgumentException("Invalid refund amount.");
        }
        System.out.printf("UPI refund: Rs. %.2f%n", refundAmount);
    }
}
```

```
package payment;

public class Main {

    public static void main(String[] args) {

        Payment card = new CardPayment(2000);
        Payment upi = new UPIPayment(1500);
        Payment cash = new CashPayment(500);

        card.pay();
        upi.pay("UPI-TXN-101");
        cash.pay(10, "CASH-101");

        Refundable refundable = (Refundable) card;
        refundable.refund(500);
    }
}
```

